

Degree- $2p$ resonance in Hénon–Frobenius dynamics

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Abstract

Let p be an odd prime, $q = p^e$ with $e \geq 3$, and let $g \in \mathbb{F}_q[y]$ have degree $2p$. For every $a \in \mathbb{F}_q^*$ we determine the ordinary geometric fixed points of every iterate of $S = H^{-1}\Phi_q$, where $H(x, y) = (y, y^q + g(y) - ax)$ and Φ_q is coordinatewise q -Frobenius. The answer has two coefficient-support branches. If g has a nonzero term of degree strictly between $p + 1$ and $2p$, a two-term degree block gives the count. If all such terms vanish, a different recurrence arises from a semilinear operator on a perfected polynomial ring. A two-term invariant controls every fractional and mixed remainder and descends to an ordinary polynomial, including arbitrary coefficients outside the prime field. Writing $w = p^{v_p(n)}$, both branches yield an explicit integer D_w and the formula $\# \text{Fix}(S^n) = q^{2n-w} D_w$. The relevant fixed schemes are finite and reduced. The resulting dynamical zeta function, and every positive integral power of it, has the circle $|t| = q^{-2}$ as a meromorphic natural boundary.

Keywords: Hénon automorphism; finite field; Frobenius; ordinary fixed points; perfection; dynamical zeta function.

1 The full coefficient-family count

In characteristic p , the first binomial term surviving a subtraction can depend on the p -valuation of an exponent. For Hénon–Frobenius fixed-point equations this makes a lower coefficient decisive even when the total degree is fixed. We determine this dependence for every degree- $2p$ correction, rather than for a prescribed sparse support.

Throughout the paper let

$$p \text{ be odd,} \quad q = p^e, \quad e \geq 3, \quad g(y) = \sum_{i=0}^{2p} b_i y^i \in \mathbb{F}_q[y], \quad b := b_{2p} \neq 0, \quad a \in \mathbb{F}_q^*. \quad (1.1)$$

Set $K = \overline{\mathbb{F}}_q$ and define maps on the whole affine plane by

$$H(x, y) = (y, y^q + g(y) - ax), \quad \Phi_q(x, y) = (x^q, y^q), \quad S = H^{-1}\Phi_q. \quad (1.2)$$

Our observable is

$$N_n = \#\{P \in \mathbb{A}^2(K) : S^n(P) = P\}, \quad n \geq 1,$$

where points are counted without multiplicity. The time variable is ordinary positive iteration of the one specified map S .

The following two alternatives partition all coefficients allowed by (1.1).

- In the *high-support class*, some $b_i \neq 0$ for $p + 2 \leq i < 2p$. Define

$$\ell = \max\{i : p + 2 \leq i < 2p, b_i \neq 0\}, \quad c = b_\ell, \quad h = 2p - \ell.$$

Thus $1 \leq h \leq p - 2$, $c \neq 0$, and $p \nmid \ell$.

- In the *low-support class*, all those coefficients vanish, and we write

$$g(y) = by^{2p} + cy^{p+1} + R_0(y), \quad \deg R_0 \leq p.$$

Here $c = b_{p+1}$ is allowed to vanish.

The notation c is local to the applicable alternative. All lower coefficients and the nonzero constants a, b remain unrestricted in $\mathbb{F}_q$.

Theorem 1.1 (All-period ordinary counts). *Under (1.1), define, for $j \geq 1$,*

$$D_j = \begin{cases} \frac{(\ell - 1)q^j + q(h + 1) - 2p}{q - 1}, & \text{in the high-support class,} \\ \frac{pq^j + (q - 2p)p^j}{q - p}, & \text{in the low-support class.} \end{cases} \quad (1.3)$$

These are positive integers with $D_j < q^j$. For every $n \geq 1$, put $w = p^{v_p(n)}$. The fixed scheme of S^n is finite and reduced, and

$$\boxed{N_n = q^{2n-w} D_w.} \quad (1.4)$$

In particular, if $p \nmid n$, then $N_n = 2p q^{2n-1}$ in both classes.

The two formulas agree at $j = 1$, but their resonant continuations are different. The mechanism behind this distinction is summarized in Table 1. The low-support argument is not obtained by setting a coefficient equal to zero in the high-support formula: the retained degree block itself changes.

Table 1: The degree states underlying Theorem 1.1. All coefficients not mentioned in the support condition are arbitrary.

Class	Retained state	Degree recurrence
High support	Two integral powers, separated by h	$D_1 = 2p,$ $D_{j+1} = q(D_j - h - 1) + 2p$
Low support	Two perfected powers, separated by $(p - 1)/p$	$E_1 = 2, E_{j+1} = (q/p)(E_j - 1) + 2,$ $D_j = p^j E_j$

1.1 Relation to existing mechanisms

Artin–Mazur zeta functions in positive characteristic already exhibit arithmetic dependence on the iterate. Bridy determines rationality or transcendence for dynamically affine rational maps on the projective line [2]. Byszewski, Cornelissen and Houben study smooth algebraic-group endomorphisms and a class of finite-adelically distorted fixed-point sequences [3]. Those are source frameworks and comparison classes; neither is used here as a theorem asserting the present nonlinear coefficient-family formula. We make no claim that all positive-characteristic zeta functions satisfy a single such dichotomy.

The internal predecessor C404 [1] supplies the commuting-pullback route from a degree calculation to reduced ordinary counts, and the positive-tail argument for a zeta natural boundary. We reproduce both arguments so that no external proof file is needed to read this

paper. The integral strict-gap calculation is likewise an input mechanism, presented in full in Section 4. The additional result here is the complete degree- $2p$ support classification for all odd p and all $q = p^e$ with $e \geq 3$. Its essential step is the low-support semilinear induction, including fractional tails and descent, in Sections 5–6. The natural boundary is a consequence of that count, not a separate new analytic mechanism.

Section 2 fixes the algebraic conventions, and Section 3 proves the conditional count conversion. The two degree calculations then prove Theorem 1.1. Section 7 gives the explicit zeta product and its boundary; Section 8 records a support-sensitive example and the limits of the result.

2 Equalizers and commuting pullbacks

The automorphism H has inverse

$$H^{-1}(x, y) = (a^{-1}(x^q + g(x) - y), x).$$

All its coefficients are fixed by q -Frobenius, so $H\Phi_q = \Phi_q H$. Consequently

$$S(x, y) = (a^{-1}(x^{q^2} + g(x^q) - y^q), x^q), \quad S^n = H^{-n}\Phi_q^n. \quad (2.1)$$

In particular S is a finite radicial map: coordinatewise Frobenius is finite and radicial, and composing with an automorphism preserves these properties. The universal-homeomorphism property of Frobenius is standard; see [5, Tag 0CC8]. Radicality does not mean that a fixed-point scheme must carry the degree of a geometric fiber as its point count.

Composing the equalizer equations with H^n identifies the fixed scheme of S^n with

$$\mathcal{E}_n = \{H^n(P) = \Phi_q^n(P)\}. \quad (2.2)$$

Equivalently, the two pairs of coordinate differences generate the same equalizer ideal after this automorphism: differences of polynomial substitutions lie in the ideal generated by the original coordinate differences, and H^{-n} gives the reverse inclusion.

Let $R = \mathbb{F}_q[x, y]$, and introduce the $\mathbb{F}_q$ -linear operators

$$T = H^*, \quad U = \Phi_q^*, \quad \delta = T - U. \quad (2.3)$$

Here pullback means substitution; T and U are ring homomorphisms, whereas their difference δ is only an additive linear operator. In particular δ^j is an operator iterate, not an iterate of a point map. The commutation above gives $TU = UT$. Moreover $U(P) = P^q$ for $P \in R$.

Write $u_0 = g(y) - ax$. On the two coordinates,

$$\delta x = y - x^q, \quad \delta y = u_0, \quad Ty = y^q + u_0, \quad (2.4)$$

and for every $P \in R$,

$$\deg \delta P \leq q \deg P. \quad (2.5)$$

We use total degree and put $\deg 0 = -\infty$. A polynomial has a *unique top term* $d_0 y^D$ if its difference from $d_0 y^D$ has total degree strictly less than D , where $d_0 \neq 0$.

The degree calculation needed for all periods is the following more precise version of (1.3):

$$\delta^j y = c_j y^{D_j} + \text{terms of degree} < D_j, \quad c_j = \begin{cases} b(c\bar{\ell})^{j-1}, & \text{high support,} \\ 2^{j-1} b^{(p^j-1)/(p-1)}, & \text{low support.} \end{cases} \quad (2.6)$$

In this expression $\bar{\ell}$ denotes the residue of ℓ in $\mathbb{F}_p^*$. All c_j are nonzero. We first explain why this one sequence of leading terms determines every ordinary fixed count.

3 From one degree sequence to every ordinary count

Proposition 3.1 (Conditional count conversion). *Suppose that for every $j \geq 1$ the polynomial $\delta^j y$ has a unique top term $c_j y^{D_j}$ with $c_j \in \mathbb{F}_q^*$ and $0 < D_j < q^j$. Then for every $n \geq 1$, with $w = p^{v_p(n)}$, the scheme $\mathcal{E}_n$ is finite and reduced and has $q^{2n-w} D_w$ points.*

Proof. Write $n = ws$, where $p \nmid s$. The characteristic- p binomial identity in the algebra of commuting linear operators gives

$$T^w - U^w = \delta^w, \quad T^n - U^n = \sum_{i=0}^{s-1} T^{wi} U^{w(s-1-i)} \delta^w. \quad (3.1)$$

Either T or U sends a polynomial with unique top term $d_0 y^D$ to one with unique top term $d_0 y^{qD}$. Indeed, the substitution of any lower-degree monomial has degree at most $q(D-1)$, whereas the leading y -monomial acquires degree qD from the monic y^q term. Both operators fix the coefficient $d_0 \in \mathbb{F}_q$. Every summand of (3.1), applied to y , therefore has the same top term. Since s is nonzero in $\mathbb{F}_q$,

$$H_2^n(x, y) - y^{q^n} = s c_w y^{q^{n-w} D_w} + \text{terms of lower total degree}. \quad (3.2)$$

For the first equation, $\deg H_1^n = q^{n-1}$. This follows by induction from $H_1^{n+1} = H_2^n$ and the unique top term y^{q^n} in H_2^n ; the latter persists because $\deg g < q$. Thus

$$H_1^n(x, y) - x^{q^n} = -x^{q^n} + \text{terms of lower total degree}. \quad (3.3)$$

Choose a monomial order refining total degree. After rescaling the two equations, their leading monomials are

$$x^M, \quad y^L, \quad M = q^n, \quad L = q^{n-w} D_w.$$

They are relatively prime, so these two polynomials form a Gröbner basis. To recall the elementary criterion in this case, write them as $F = x^M + F_0$ and $G = y^L + G_0$, with $\deg F_0 < M$ and $\deg G_0 < L$. Their sole S -polynomial has the standard representation

$$y^L F - x^M G = F_0 G - G_0 F,$$

whose summands have leading monomials strictly below $x^M y^L$. Buchberger's criterion applies. This is the standard argument for coprime leading monomials; see [4] for the general framework. Consequently the quotient over K has basis

$$\{x^u y^v : 0 \leq u < M, 0 \leq v < L\} \quad (3.4)$$

and dimension $ML = q^{2n-w} D_w$. This proves finiteness and the scheme length. It also shows directly that no projective point at infinity contributes: the leading forms x^M, y^L have no common zero on the line at infinity.

It remains to justify passage from length to ordinary points. The Jacobian of the defining pair in (2.2) is DH^n , since derivatives of q^n -powers vanish in characteristic p . As $\det DH = a$, its determinant is $a^n \neq 0$ at every point. At each solution, the equations have independent linear parts in the ambient two-dimensional cotangent space. The cotangent space of the quotient local Artinian algebra is therefore zero: its maximal ideal modulo its square vanishes. Nakayama's lemma makes that maximal ideal zero. The local algebra is K , and every point has length one. Thus $\mathcal{E}_n$ is reduced and its length is its ordinary geometric count. $\square$

The argument is independent of how the degree sequence is obtained. The coefficient-support distinction enters only in establishing (2.6), to which we now turn.

4 The integral two-term block in the high-support class

Assume throughout this section that $g = by^{2p} + cy^\ell + R_1(y)$, where $\deg R_1 < \ell$, $p + 2 \leq \ell < 2p$, and $b, c \neq 0$. No multiple of p lies strictly between ℓ and $2p$. With $h = 2p - \ell$, the hypotheses $q \geq p^3$ and $1 \leq h \leq p - 2$ imply the two strict inequalities

$$q - 2p > h, \quad q(p - h - 1) - 2p(p - 1) > h. \quad (4.1)$$

For the second, $p - h - 1 \geq 1$ and

$$q(p - h - 1) - 2p(p - 1) \geq p^3 - 2p(p - 1) > p - 2 \geq h.$$

These margins are uniform over the entire high-support coefficient class, including $q = p^3$.

Lemma 4.1 (First nonzero binomial index). *If $D \geq 1$ and $r_D = p^{v_p(D)}$, then*

$$\deg((y^q + u_0)^D - y^{qD}) \leq q(D - r_D) + 2pr_D. \quad (4.2)$$

The first potentially nonzero binomial index is r_D .

Proof. Write $D = r_D d$ with $p \nmid d$. The identity $(Y + Z)^D = (Y^{r_D} + Z^{r_D})^d$ shows that the first nonconstant term is $dY^{r_D(d-1)}Z^{r_D}$. Since $\deg u_0 = 2p < q$, a term with a larger index has strictly smaller total degree than the one with index r_D . This proves the bound. $\square$

Proposition 4.2 (High-support invariant). *Define integers and nonzero coefficients by*

$$A_1 = 0, \quad A_{j+1} = q(A_j + \ell - 1), \quad C_j = (c\bar{\ell})^{j-1}.$$

Then for every $j \geq 1$,

$$\delta^j y = C_j y^{A_j} (by^{2p} + cy^\ell) + R_j, \quad \deg R_j < A_j + \ell. \quad (4.3)$$

In particular (2.6) holds with the high-support degree in (1.3).

Proof. At $j = 1$, the assertion is $\delta y = g(y) - ax$; the omitted terms have degree below ℓ . Suppose (4.3) holds at j , and set

$$D = A_j + 2p, \quad E = A_j + \ell, \quad d = q(E - 1) + 2p.$$

The integer A_j is zero or divisible by q . Because $2p < q$, $v_p(D) = 1$, whereas $E \equiv \ell \pmod{p}$ is nonzero.

Consider first $C_j c \delta(y^E)$. Its binomial-index-one term is

$$C_j c \bar{\ell} y^{q(E-1)} (g(y) - ax).$$

This supplies precisely the two displayed terms at degrees d and $d - h$, with common multiplier C_{j+1} . Every other term in this index-one contribution has smaller degree than $d - h$. Each term with index $i \geq 2$ has degree at most

$$q(E - i) + 2pi = d - (i - 1)(q - 2p) < d - h$$

by the first inequality in (4.1).

For the leading monomial $C_j b \delta(y^D)$, Lemma 4.1 gives degree at most $q(D - p) + 2p^2$. Its deficit from d is

$$d - (q(D - p) + 2p^2) = q(p - h - 1) - 2p(p - 1) > h.$$

Thus this term changes neither retained coefficient. Finally, (2.5) and the integral degree of R_j give

$$\deg \delta R_j \leq q(E-1) = d - 2p < d - h.$$

All omitted terms lie below the second retained term. This proves (4.3) at $j+1$.

The top degree is $D_j = A_j + 2p$, and hence

$$D_1 = 2p, \quad D_{j+1} = q(D_j - h - 1) + 2p.$$

Solving this affine recurrence yields the first formula in (1.3). Its coefficient is bC_j , as in (2.6). Positivity and integrality follow from $A_j \geq 0$. Also $D_1 < q$ and $D_{j+1} < qD_j$, since $q(h+1) > 2p$, so $D_j < q^j$ for every j . $\square$

The proof uses strict margins to prevent a lower term from entering the retained block. Those margins no longer apply at the low-support boundary $\ell = p+1$. A different state is needed there, including when the coefficient of y^{p+1} vanishes altogether.

5 A semilinear operator on a perfected polynomial ring

We now assume the low-support form

$$g(y) = by^{2p} + cy^{p+1} + R_0(y), \quad \deg R_0 \leq p, \quad b \neq 0, \quad (5.1)$$

with no condition on c . The proof will take roots of polynomial expressions, but only in the following concrete algebraic domain:

$$\mathcal{R} = \bigcup_{r \geq 0} \mathbb{F}_q[x^{1/p^r}, y^{1/p^r}]. \quad (5.2)$$

Every element is a finite sum of monomials with nonnegative exponents in $\mathbb{Z}[1/p]$. No formal infinite series or completed ring is involved. The monomial representation is unique, and rational total degree satisfies

$$\deg(PQ) = \deg P + \deg Q, \quad \deg(P^{1/p}) = \frac{1}{p} \deg P$$

for nonzero P, Q . The ordinary ring R embeds injectively.

Let L be absolute Frobenius on $\mathcal{R}$, so $L(P) = P^p$. This is a ring automorphism, whose inverse acts on coefficients by

$$\sigma(d) = d^{1/p} = d^{p^{e-1}} \quad (d \in \mathbb{F}_q). \quad (5.3)$$

The maps T, U extend uniquely from R to $\mathcal{R}$: evaluate an ordinary polynomial and then take its unique prescribed root. They commute with L and L^{-1} . In addition,

$$\delta(P^p) = (TP)^p - (UP)^p = (\delta P)^p.$$

Thus δ also commutes with L . Define the additive operator

$$\mathcal{D} = L^{-1}\delta. \quad (5.4)$$

The two identities that will control both degrees and coefficients are

$$\delta^j = L^j \mathcal{D}^j, \quad \mathcal{D}(dP) = \sigma(d) \mathcal{D}(P) \quad (d \in \mathbb{F}_q). \quad (5.5)$$

The second is semilinearity, not $\mathbb{F}_q$ -linearity. It cannot be discarded when a, b, c lie outside the prime field. The first is an identity of additive operators, not a conjugacy of point maps and not a change of the time variable n .

Put

$$Q = q/p, \quad B = \sigma(b) \neq 0, \quad C = \sigma(c), \quad \varepsilon = (p-1)/p. \quad (5.6)$$

The maps $V = L^{-1}T$ and $W = L^{-1}U$ are ring homomorphisms with coefficient action σ , and $\mathcal{D} = V - W$. Their coordinate substitutions are

$$Vx = y^{1/p}, \quad Wx = x^Q, \quad Vy = y^Q + v_0, \quad Wy = y^Q, \quad (5.7)$$

where

$$v_0 = By^2 + Cy^{1+1/p} + v_1, \quad \deg v_1 \leq 1. \quad (5.8)$$

Indeed, every term of R_0 has degree at most p before taking a root, while $-ax$ becomes $-\sigma(a)x^{1/p}$. This also shows explicitly why arbitrary low support and the determinant term fit into the same tail bound.

Lemma 5.1 (Uniform mixed-monomial bound). *For every $P \in \mathcal{R}$,*

$$\deg \mathcal{D}P \leq Q \deg P. \quad (5.9)$$

The bound is independent of the denominator level of P .

Proof. Take a mixed monomial $x^u y^v$ with $u, v \in \mathbb{Z}[1/p]_{\geq 0}$, and choose a common denominator p^r . Its image under each substitution can be computed as the unique p^r th root of the corresponding ordinary-power expression. Since $Q > 2$, (5.7)–(5.8) give

$$\deg V(x^u y^v) = u/p + Qv \leq Q(u + v), \quad \deg W(x^u y^v) = Q(u + v).$$

For a finite sum, cancellations can only decrease degree. The coefficient automorphism leaves a nonzero scalar of degree zero, so the same bound holds for $V - W$ and every P . $\square$

This estimate is deliberately uniform. The subsequent induction will produce fractional exponents and mixed x, y terms; it does not assume that all omitted terms remain univariate or share a fixed denominator.

6 The low-support invariant and its descent

The perfected operator lowers the first degree from $2p$ to 2. We retain the next possible exponent as well, because its denominator changes the first surviving binomial index.

Lemma 6.1 (Integral and exceptional fractional exponents). *Let $E \geq 2$ be an integer with $E \equiv 2 \pmod{p}$, and set $E' = Q(E - 1) + 2$. Then*

$$\mathcal{D}(y^E) = 2By^{E'} + 2Cy^{E'-\varepsilon} + P_E, \quad \deg P_E \leq E' - 1, \quad (6.1)$$

whereas

$$\deg \mathcal{D}(y^{E-\varepsilon}) = Q(E - 1) + 2/p < E' - 1. \quad (6.2)$$

Proof. For the integral exponent, use the finite expansion $(y^Q + v_0)^E - y^{QE}$. Its index-one term is $2y^{Q(E-1)}v_0$, because $E \equiv 2 \pmod{p}$. Formula (5.8) gives the two terms displayed in (6.1); the remainder of that index-one term has degree at most $E' - 1$. A term with index $i \geq 2$ has degree at most

$$Q(E - i) + 2i = E' - (i - 1)(Q - 2) \leq E' - 1.$$

Here $Q \geq p^2 \geq 9$, and 2 is nonzero in the odd characteristic.

For the fractional exponent write

$$E - \varepsilon = \frac{M}{p}, \quad M = p(E - 1) + 1.$$

The ring-homomorphism property and uniqueness of roots give the exact finite identity

$$\mathcal{D}(y^{E-\varepsilon}) = ((y^Q + v_0)^M - y^{QM})^{1/p}. \quad (6.3)$$

Although v_0 itself may have fractional exponents, the expression inside the root is a finite binomial expansion in $\mathcal{R}$. Since $M \equiv 1 \pmod{p}$, its index-one term has degree $Q(M - 1) + 2$, with nonzero leading coefficient B . Every larger index has strictly smaller degree, as $Q > 2$. Taking the root divides this degree by p , so

$$\deg \mathcal{D}(y^{E-\varepsilon}) = \frac{Q(M - 1) + 2}{p} = Q(E - 1) + \frac{2}{p} = E' - \left(2 - \frac{2}{p}\right).$$

For every odd p , including $p = 3$, one has $2 - 2/p > 1$. This proves (6.2). $\square$

Proposition 6.2 (Low-support state). *For every $j \geq 1$ there are $\alpha_j, \beta_j \in \mathbb{F}_q$, $\alpha_j \neq 0$, and $P_j \in \mathcal{R}$ such that*

$$\mathcal{D}^j y = \alpha_j y^{E_j} + \beta_j y^{E_j - \varepsilon} + P_j, \quad \deg P_j \leq E_j - 1. \quad (6.4)$$

They satisfy

$$E_1 = 2, \quad E_{j+1} = Q(E_j - 1) + 2, \quad (6.5)$$

$$\alpha_1 = B, \quad \alpha_{j+1} = 2B\sigma(\alpha_j), \quad (6.6)$$

$$\beta_1 = C, \quad \beta_{j+1} = 2C\sigma(\alpha_j). \quad (6.7)$$

Proof. At $j = 1$ the invariant is (5.8), since $2 - \varepsilon = 1 + 1/p$. The recurrence for E_j makes it an integer at least two with $E_j \equiv 2 \pmod{p}$, because $p \mid Q$. Assume the invariant at j . Apply semilinearity (5.5) and Lemma 6.1 to the first two terms. The leading term supplies exactly the two coefficients in (6.6)–(6.7). The exceptional fractional term has degree strictly below $E_{j+1} - 1$ after applying $\mathcal{D}$. For the entire remaining tail, Lemma 5.1 gives

$$\deg \mathcal{D}P_j \leq Q(E_j - 1) = E_{j+1} - 2.$$

Thus no omitted term can enter either retained position, and (6.4) follows at $j + 1$. Since $B \neq 0$, $2 \neq 0$, and σ is an automorphism, α_j never vanishes. If $C = 0$, all β_j vanish; no division by C has been made and every bound remains valid. $\square$

Proposition 6.3 (Ordinary polynomial descent). *For the low-support class, $\delta^j y$ has the unique top term in (2.6), with the low-support degree in (1.3). In particular $0 < D_j < q^j$.*

Proof. Solving (6.5) gives

$$E_j = \frac{Q^j + Q - 2}{Q - 1}. \quad (6.8)$$

Each application of $\mathcal{D} = L^{-1}\delta$ raises the denominator level by at most one, since T, U preserve each level of (5.2). Hence $\mathcal{D}^j y$ is at level at most j . Taking its p^j th power gives, by (5.5), the ordinary polynomial $\delta^j y$.

This descent can also be seen term by term in the retained state. The top exponent becomes $D_j = p^j E_j$; the second becomes

$$p^j(E_j - \varepsilon) = D_j - (p - 1)p^{j-1},$$

an integer strictly below it. The descended tail has degree at most $D_j - p^j$. Thus the top term remains unique; neither a fractional exponent nor a possible mixed remainder has been left unresolved. Substituting $Q = q/p$ into (6.8) gives

$$D_j = p^j E_j = \frac{pq^j + (q - 2p)p^j}{q - p}.$$

This is a positive integer by the recurrence. Moreover $E_1 = 2 < Q$ and $E_{j+1} = QE_j - (Q - 2) < QE_j$, so $E_j < Q^j$ and $D_j < q^j$.

Finally the top coefficient is $c_j = \alpha_j^{p^j}$. From (6.6), using $\sigma(\alpha_j)^p = \alpha_j$ and $B^p = b$, we obtain

$$c_1 = b, \quad c_{j+1} = 2b^{p^j} c_j.$$

Therefore

$$c_j = 2^{j-1} b^{1+p+\dots+p^{j-1}} = 2^{j-1} b^{(p^j-1)/(p-1)}.$$

This accounts for all coefficient Frobenius twists, without requiring any coefficient to lie in $\mathbb{F}_p$. $\square$

Proof of Theorem 1.1. Proposition 4.2 proves the needed leading terms and degree inequalities in the high-support class; Proposition 6.3 proves them in the low-support class. These alternatives exhaust every degree- $2p$ polynomial in (1.1). Proposition 3.1, together with the fixed-scheme identification (2.2), proves the ordinary count and reducedness. At $j = 1$ both degree formulas equal $2p$, giving the nonresonant statement when $w = 1$. $\square$

7 The native zeta product and its natural boundary

Define the dynamical zeta function for the same map and clock by

$$Z_S(t) = \exp \left(\sum_{n \geq 1} \frac{N_n}{n} t^n \right). \quad (7.1)$$

No Euler factors from another object enter this definition. For the applicable support class put

$$(A, B_0, \theta) = \begin{cases} \left(\frac{\ell-1}{q-1}, \frac{q(h+1)-2p}{q-1}, \frac{1}{q} \right), & \text{high support,} \\ \left(\frac{p}{q-p}, \frac{q-2p}{q-p}, \frac{p}{q} \right), & \text{low support.} \end{cases} \quad (7.2)$$

We use B_0 here to distinguish this real constant from the coefficient $B = \sigma(b)$ in the perfected-ring proof. Theorem 1.1 is equivalent to

$$\frac{N_n}{q^{2n}} = A + B_0 \theta^{p^{v_p(n)}}. \quad (7.3)$$

In either class,

$$A, B_0 > 0, \quad 0 < \theta < 1, \quad A + B_0 \theta = \frac{2p}{q}. \quad (7.4)$$

In particular the normalization used below is $A + B_0 \theta$, not an assumed identity $A + B_0 = 1$.

Theorem 7.1 (Product and meromorphic natural boundary). *For $|u| < 1$,*

$$Z_S(u/q^2) = (1-u)^{-2p/q} \prod_{k \geq 1} (1-u^{p^k})^{e_k}, \quad e_k = \frac{B_0}{p^k} (\theta^{p^{k-1}} - \theta^{p^k}) > 0. \quad (7.5)$$

The product is defined by logarithms equal to zero at $u = 0$ and converges locally uniformly in the unit disk. For every positive integer m , the circle $|t| = q^{-2}$ is a meromorphic natural boundary for $Z_S(t)^m$: no arc admits meromorphic continuation. In particular Z_S has radius of convergence exactly q^{-2} and is transcendental over $\mathbb{C}(t)$.

Proof. For $w = p^{v_p(n)}$ the finite telescoping identity is

$$\theta^w = \theta + \sum_{k \geq 1} (\theta^{p^k} - \theta^{p^{k-1}}) \mathbf{1}_{p^k | n}. \quad (7.6)$$

For each fixed n only finitely many summands occur. Interchanging this sum with $\sum u^n/n$ is valid uniformly on $|u| \leq r < 1$: the sum of the absolute coefficient differences over k is θ , so an upper bound for the resulting double absolute sum is $\theta \sum_{n \geq 1} r^n/n < \infty$. Also

$$\sum_{\substack{n \geq 1 \\ p^k | n}} \frac{u^n}{n} = -\frac{1}{p^k} \log(1 - u^{p^k}).$$

Substitution of (7.6) into (7.3) and (7.1) proves (7.5), with locally uniform logarithmic convergence. The resulting holomorphic function has no zeros in the unit disk, as is also immediate from (7.1).

Let ξ be a primitive p^r th root of unity, where $r \geq 1$. The restriction $r \geq 1$ ensures $\xi \neq 1$, so the initial factor $1 - u$ does not vanish along $u = \rho\xi$ as $\rho \uparrow 1$. The finitely many factors with $k < r$ likewise have nonzero limits. For $k \geq r$, $\xi^{p^k} = 1$, and

$$\log |1 - (\rho\xi)^{p^k}| = \log(1 - \rho^{p^k}).$$

For $0 < \rho < 1$,

$$0 \leq \frac{\log(1 - \rho^{p^k})}{\log(1 - \rho)} \leq 1, \quad \lim_{\rho \uparrow 1} \frac{\log(1 - \rho^{p^k})}{\log(1 - \rho)} = 1.$$

The bound $\sum e_k < \infty$ permits dominated convergence. Writing $F(u) = Z_S(u/q^2)$ gives the radial logarithmic order

$$\lim_{\rho \uparrow 1} \frac{\log |F(\rho\xi)|}{\log(1 - \rho)} = \eta_r := \sum_{k \geq r} e_k > 0. \quad (7.7)$$

These positive tails tend to zero; more explicitly,

$$0 < \eta_r \leq \frac{B_0}{p^r} \sum_{k \geq r} (\theta^{p^{k-1}} - \theta^{p^k}) = \frac{B_0}{p^r} \theta^{p^{r-1}}. \quad (7.8)$$

Fix a positive integer m . For all sufficiently large r , $0 < m\eta_r < 1$. A nonzero meromorphic germ at ξ has the form $(u - \xi)^j G(u)$ with $j \in \mathbb{Z}$ and G holomorphic and nonzero at ξ . Its radial logarithmic order is the integer j . Equation (7.7) makes that impossible for F^m at every primitive p^r th root with r sufficiently large. Such roots are dense on the unit circle, so no open arc can admit meromorphic continuation of F^m .

The defining series is holomorphic in $|u| < 1$ by (7.3); the boundary just established precludes a larger disk. An algebraic function has only finitely many singular or branch points in the finite plane and therefore has continuation across some arc of this circle. Thus F , and equivalently Z_S , is not algebraic over $\mathbb{C}(t)$. $\square$

This proof is the positive-tail mechanism already used in C404 [1]. The contribution of the present degree calculation is to determine which of the two exact triples in (7.2) governs every coefficient choice.

8 Support sensitivity and the limits of the theorem

Example 8.1 (The smallest field boundary). Take $p = 3$, $q = 27$, and $a = 1$. Both $g_0(y) = y^6$ and $g_1(y) = y^6 + y^5$ have degree six, but their third fixed counts differ. The first is low support and the second has $\ell = 5$, $h = 1$. Substitution into the proved formulas yields

g	D_3	$N_3 = 27^3 D_3$
y^6	2484	48 892 572
$y^6 + y^5$	3030	59 639 490

These numbers are evaluations of Theorem 1.1, not a finite-field census used to establish its quantifiers. They show why total degree alone does not determine the resonant count.

The theorem includes the pure p th-power support face, arbitrary constant term, $q = p^3$, and all coefficients in $\mathbb{F}_q$, not just $\mathbb{F}_p$. The high-support result is independent of every coefficient below y^ℓ ; the low-support result is independent of every coefficient below the leading y^{2p} for purposes of the final count. That independence follows from the proved degree estimates, not from replacing these coefficients by zero.

Several boundaries of the statement are essential to its present proof. Odd characteristic keeps the leading factor 2 nonzero and makes the fractional gap in (6.2) strict. The assumption $e \geq 3$ gives the uniform high-support margins and the simple low-support estimates. We do not assert either formula for $p = 2$ or $e = 1, 2$. Nor do we treat $a = 0$, a vanishing leading coefficient, correction degrees other than $2p$, or coefficients not fixed by q -Frobenius. The last change would remove the commutation used in (3.1).

The observable is the ordinary geometric fixed-point count for S on $\mathbb{A}^2(K)$. It is not a formula for forward periodic points of an arbitrary separable polynomial, a count in one fixed finite extension, or a scheme-length convention with multiplicities left in place. Reducedness was proved before assigning the ordinary count. The zeta function is native to this same map and clock; no identification with a prescribed arithmetic Euler product, target zero set, or spectral realization is claimed.

The result therefore consists of one full coefficient-family classification and its native analytic consequence. The calculation in the perfected ring uses only finite algebraic expressions, with each iterate lying at a finite denominator level. Its descent returns the calculation to ordinary polynomials and ordinary points.

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